

Integrated Service Digital Network (ISDN)

ISDN is a Digital communication system used for voice, data, video and fax.

Main Channels

⇒ B Channel :- Carries voice/data, 64 Kbps.

⇒ D Channel :- Carries signaling, 16/64 Kbps.

⇒ H Channel :- High speed data channel.

BRI = 2B + D

Here, $2 \times 64 \text{ Kbps} = 128 \text{ Kbps}$
 $1 \times 16 \text{ Kbps} = 16 \text{ Kbps}$.

∴ User Rate = $128 + 16 = 144 \text{ Kbps}$

and Line Rate = $128 + 64 = 192 \text{ Kbps}$.

PRI

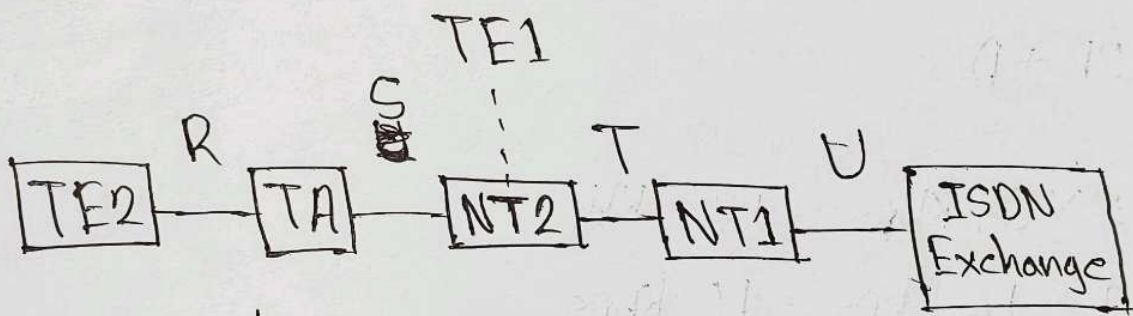
T1 = $23B + D = 1.544 \text{ Mbps}$

E1 = $30B + D = 2.048 \text{ Mbps}$

ISDN Architecture

- ⇒ TE1 (Terminal Equipment 1) → ISDN Device.
- ⇒ TE2 (Terminal Equipment 2) → Non ISDN Device.
- ⇒ TA (Terminal Adapter) → Connects TE2 to ISDN network.
- ⇒ NT1 (Network Termination 1) → Connects user side to ~~network~~.
- ⇒ NT2 (Network Termination 2) → Local switching device.

Reference Points



R → Connects TE2 - TA

S → TA/TE1 - NT2

T → NT2 - NT1

U → NT1 - Exchange

ISDN Service

1. Bearer :- Carries Data.
2. Teleservice :- Complete service like telephone/fax.
3. Supplementary :- Extra features like call waiting.

~~Advantages~~:-

ISDN Protocols:-

⇒ D Channel

Layer 1: 1.430/1.431

Layer 2: Q.921/LAPD

Layer 3: Q.931 (Handles call control messages)

⇒ B Channel → It mainly carries user data. ISDN define only the physical layer for it. Higher level protocols can be used above it.

Advantages:-

1. End to end digital communication.

2. Low noise.

3. Low error.

4. Multiple services ~~for~~ ^{on} one line.

5. Fast call setup.

6. Good signaling features.

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Mylastin
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Limitations :-

1. Low speed
2. Costly for always on internet.
3. Expensive and complex.
4. Replaced by DSL/Fiber

BRI VS PRI

BRI	PRI
1. $BRI = 2B + D$	1. $PRI = 23B + D / 30B + D$
2. For small users.	2. For Business
3. 192 Kbps line.	3. 1.544 / 2.048 Mbps

Broadband ISDN (B-ISDN) vs Narrowband ISDN (N-ISDN)

N-ISDN	B-ISDN
1. Speed 144 Kbps - 2 Mbps	1. Speed 155/622 Mbps and beyond
2. Uses circuit switching.	2. Uses ATM-based fast packet switching.
3. Uses fixed 64-Kbps channels.	3. Uses fixed 53-byte cells.
4. Mainly copper local loop.	4. Mainly optical fiber.
5. Voice & modest data.	5. Video & multimedia.

Open Systems Interconnection (OSI)

It is a reference model, used to explain how data moves from one computer to another through a network.

It has 7 layers.

1. Application layer :-

PDU $\Rightarrow$ Data

$\Rightarrow$ Closest to the user.

$\Rightarrow$ Provides network services to application.

$\Rightarrow$ e.g., web, mail, file transfer.

2. Presentation layer:-

PDU $\rightarrow$ Data

$\Rightarrow$ Handles data format.

$\Rightarrow$ Translation

$\Rightarrow$ Encryption / Decryption.

$\Rightarrow$ Compression.

3. Session layer :-

PDU $\rightarrow$ Data

$\Rightarrow$ Starts a session.

$\Rightarrow$ Maintains the session.

$\Rightarrow$ Ends the session.

4. Transport layer :-

PDU → ~~Packet~~ Segment

- ⇒ Provides end to end delivery.
- ⇒ Break data into smaller parts.
- ⇒ Error and flow control.
- ⇒ TCP and UDP works.

5. Network layer :- Device → Router, PDU → Packet

- ⇒ Provides logical addressing.
- ⇒ Finds the best path.
- ⇒ Routing is ~~no~~ done and IP works here.

6. Data link layer :- Device → Switch, bridge, PDU → Frame

- ⇒ Provides node to node delivery.
- ⇒ Uses MAC address.
- ⇒ Error Detection.
- ⇒ Creates frames.

7. Physical layer :- Device → Hub, Repeater, PDU → Bits

- ⇒ Sends raw bits through the medium.
- ⇒ Deals with cables, signals, connectors, & voltage

Open System Interconnection (OSI)

It is a reference model used to explain how data moves from one computer to another through a network.

Layer	Main Work	PDU
Application	— User Services	— Data
Presentation	— Format, encryption	— Data
Session	— Starts/manages session	— Data
Transport	— End to end delivery	— Segment
Network	— Routing, IP	— Packet
Data link	— Mac, error check	— Frame
Physical	— Sends bit	— Bits

Important Devices:-

⇒ Hub/Repeater → Physical (Layer 1)

⇒ Switch/Bridge → Data link (Layer 2)

⇒ Router → Network (Layer 3)

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~~# OSI vs TCP/IP~~

TCP/IP Model :- It is a networking model like

OSI. It has commonly 4 layers.

1. Application

2. Transport

3. Internet

4. Network Access

OSI vs TCP/IP

OSI

1. It has 7 layers.
2. Reference model.
3. Session and Presentation are separate.
4. Used for Networking.
5. Data link + Physical

TCP/IP

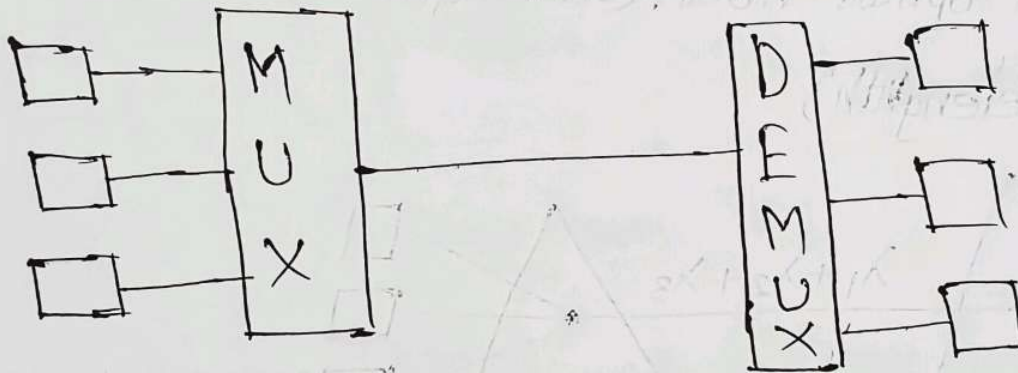
1. It has 4 layers.
2. Practical protocol model.
3. ~~Session~~ Session and presentation model included in application.
4. Used for Internet communication.
5. Network Access.

Multiplexing

It is a technique that allows multiple signals to use one communication link at the same time.

MUX (Multiplexer) → Combine signals into one.

DEMUX (Demultiplexer) → Separates signals.



Types of multiplexing :-

1. FDM

2. WDM

3. TDM → 1. Synchronous TDM

2. Asynchronous TDM

FDM:- It's an analog technique. It divides the channel according to frequency.

⇒ Applications: AM Radio, FM Radio, TV etc.

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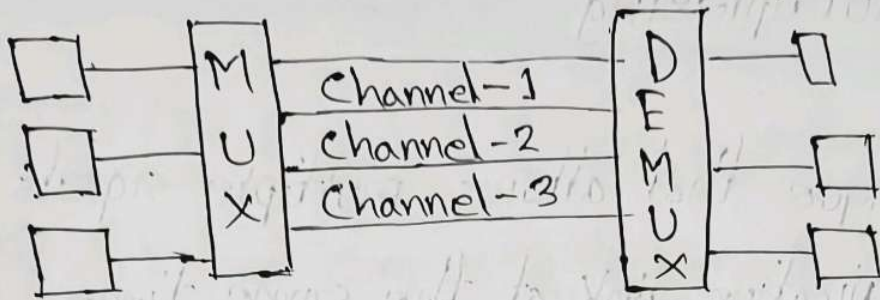


fig- FDM

WDM:- It's a analog technique. which is used with optical fiber. (diff. light signal use diff. wavelength)

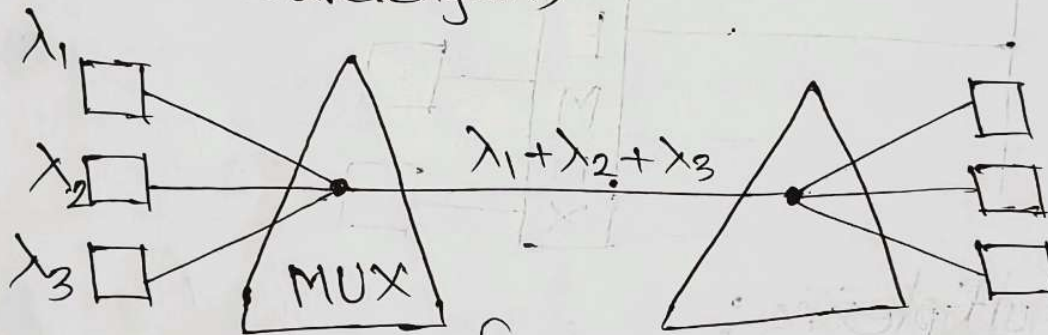


fig: WDM

TDM:- It's a digital technique. It divides the channel according to time slots.

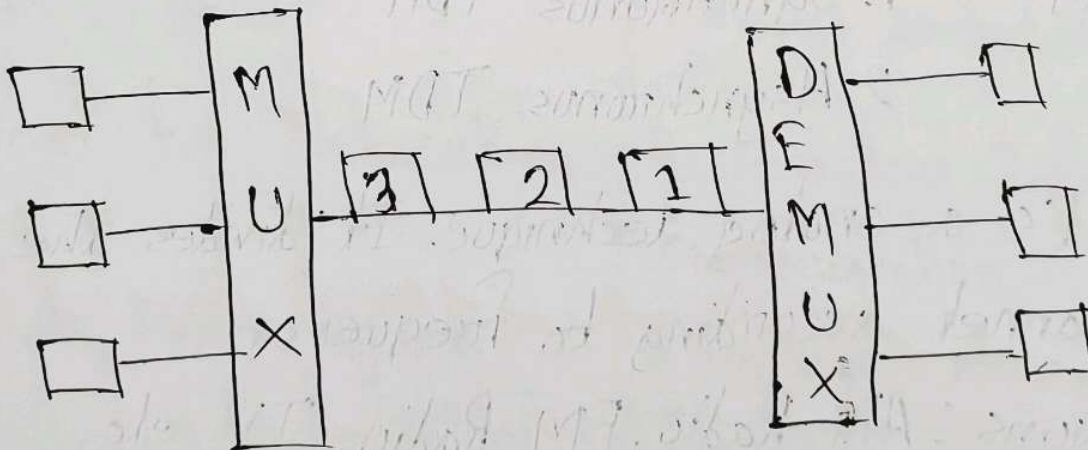


fig: TDM

1. Synchronous TDM:- Each device gets a fixed time slot, even if a device has no data, slot may remain ^{empty}.
Disadvantages → Empty slot waste channel capacity.

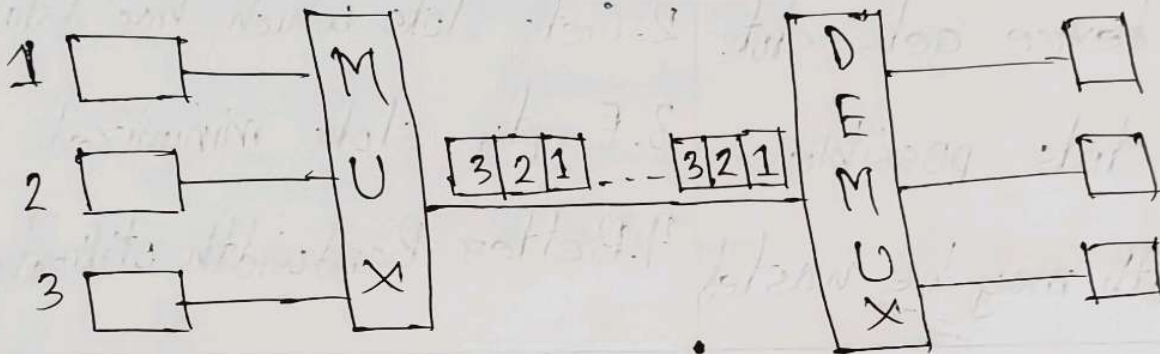


Fig: Synchronous TDM

2. Asynchronous TDM:- Time slots are flexible, slots are given to device that have data.

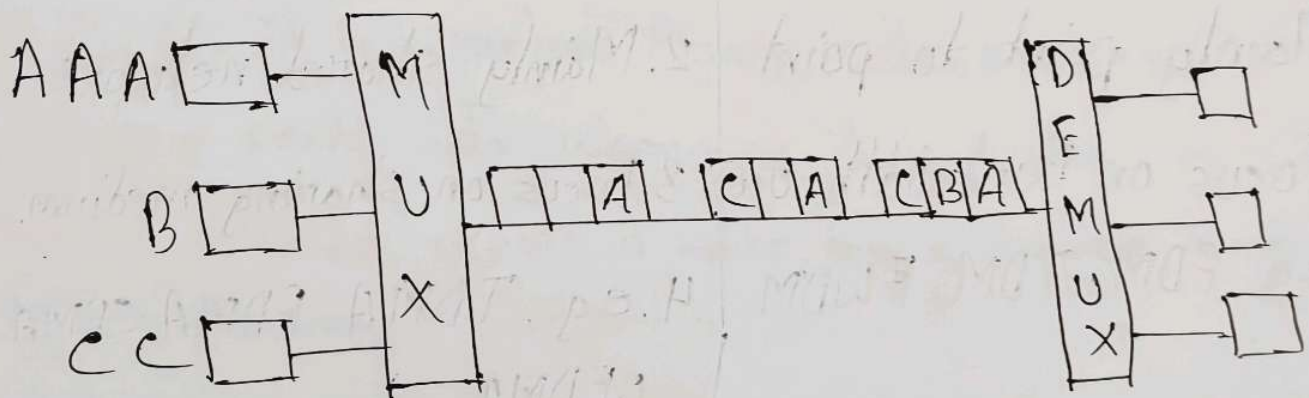


Fig: Asynchronous TDM

Synchronous TDM vs Asynchronous TDM

Synchronous TDM	Asynchronous TDM
1. Fixed Slot 2. Every device gets slot. 3. Empty slots possible 4. Bandwidth may be wasted	1. Flexible slot 2. Gets slots which has data 3. Empty slots minimized 4. Better Bandwidth utilization

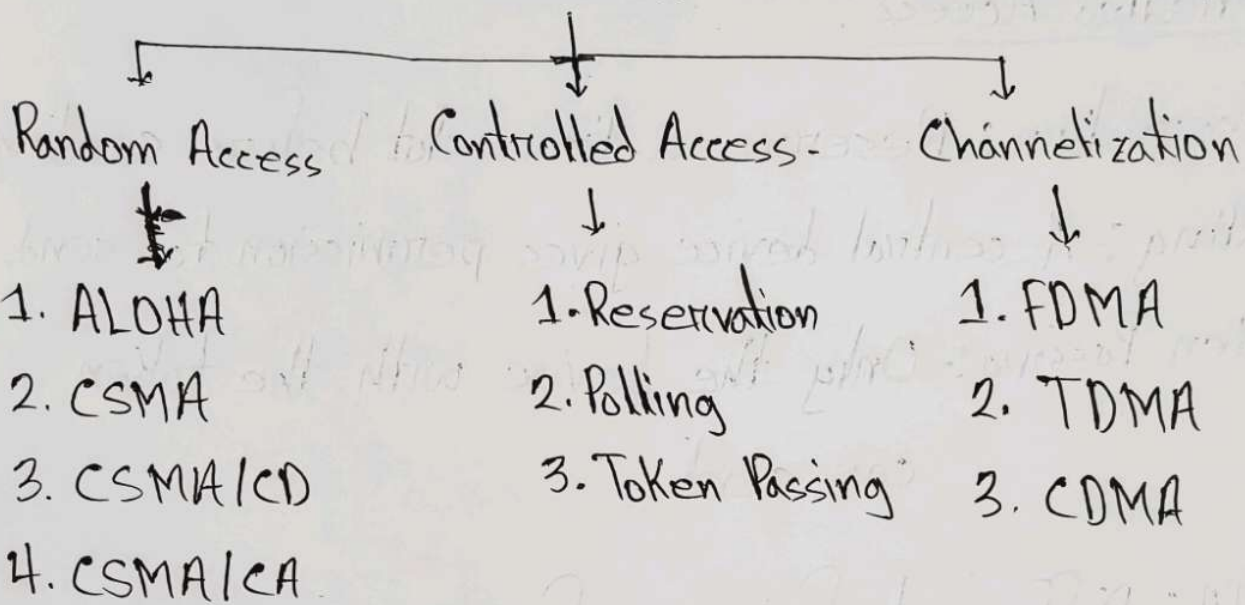
Multiplexing vs Multiple Access

Multiplexing	Multiple Access
1. Combines multiple signal. 2. Mainly point to point 3. Focus on Bandwidth use 4. e.g., FDM, TDM, WDM	1. Allows multiple users. 2. Mainly shared network 3. Focus on sharing medium. 4. e.g., TDMA, FDMA, CDMA, OFDMA

Multiple Access

It allows many users to share the same communication medium.

Multiple Access Protocols



ALOHA:- It is a random Access method where a device sends data whenever it is ready.

=> If collision occurs, it waits for a random time and sends again.

Slotted ALOHA:- It divides times into equal slots. A device can send data only at the beginning of a slot.

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⇒ It has fewer collisions than ALOHA.

CSMA:- It checks the channel before sending data.

If the channel is free, it sends.

If the channel is Busy, it waits.

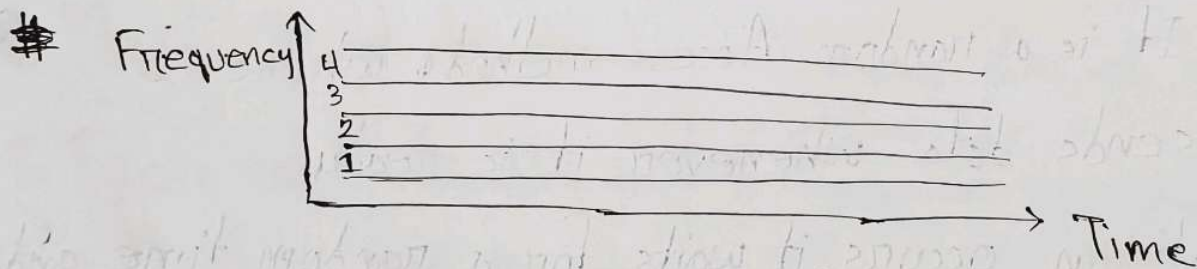
Controlled Access:-

⇒ Reservation:- Reserve a time slot before sending.

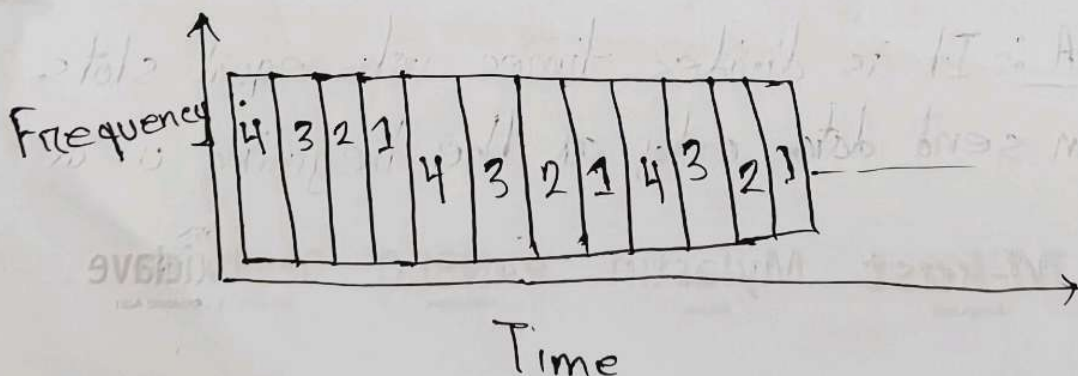
⇒ Polling:- A central device gives permission to send.

⇒ Token Passing:- Only the device with the token can send.

FDMA:- Different frequency for each user.



TDMA:- Different time slot for each user.



CDMA:- Same frequency/time but different code.

OFDMA:- User get different orthogonal sub-carriers.

Signaling System 7 (SS7)

Signaling means exchange of information between network elements. It follows protocol.

SS7:- It is a signaling system used in telecommunication networks, which was developed by ITU-T, formerly CCITT.

=> Its main idea is to send signaling information through a separate channel, instead of using the voice channel. It's called Out of band signaling or common channel signaling (CCS).

Advantages of out of band signaling:-

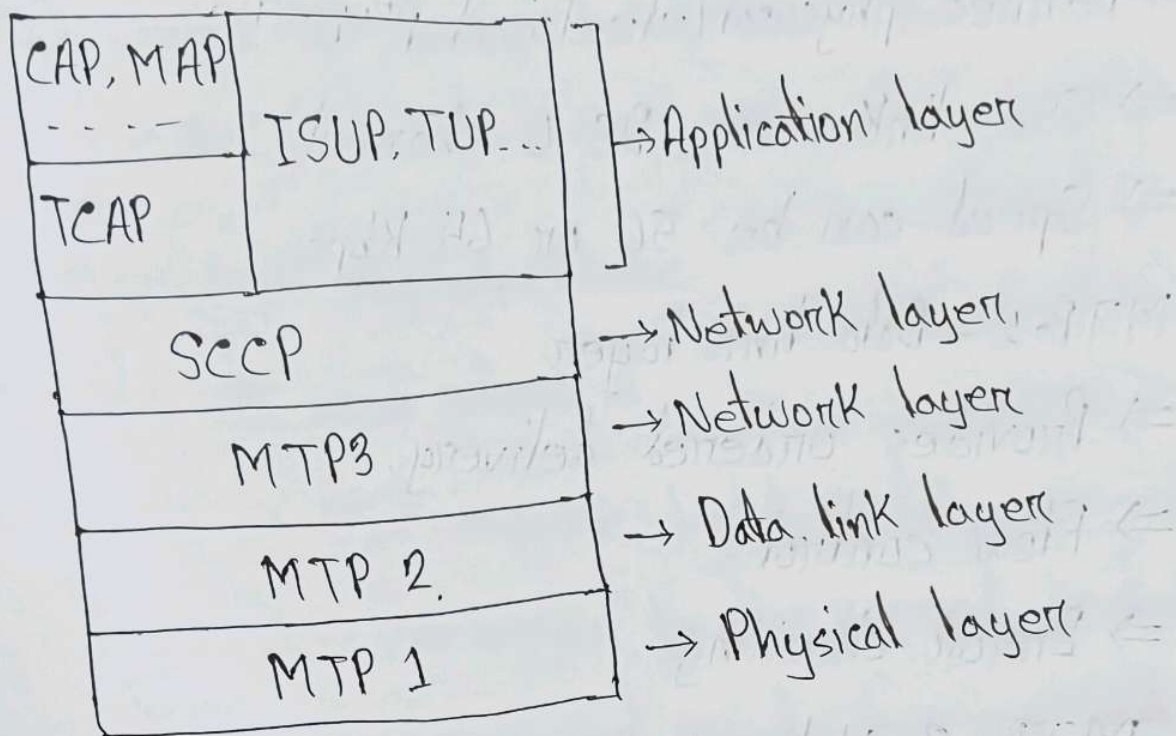
① Signaling can happen at any time.

② More signaling information can be sent.

③ Voice channel is not used for signaling.

~~# SS7 Stack and~~

SS7 Stack



SS7 Layers :-

- ⇒ MTP - Message transfer ~~protocol~~ part
- ⇒ SCCP - Signaling and connection control part
- ⇒ TCAP - Transaction Capabilities Application Part.
- ⇒ CAP - CAMEL Application Part.
- ⇒ ISUP - ISDN User Part
- ⇒ MAP - Mobile Application Part.

① MTP 1 - Physical layer

⇒ Defines physical/electrical properties.

⇒ SS7 links use DS-0 channel.

⇒ Speed can be 56 or 64 Kbps.

② MTP 2 - Data link layer

⇒ Provides ordered delivery.

⇒ Flow control

⇒ Error checking.

③ MTP 3 - Network layer

⇒ Delivers messages between signaling points.

⇒ Provides routing and message distribution.

④ SCCP - Signaling Connection Control Part.

⇒ Provides higher level end-to-end routine

⇒ Helps route TCAP messages to the correct database.

⇒ Performs Global Title translation.

⑤ TCAP - Transaction Capabilities Application Part.

⇒ Helps connect to external databases.

⇒ Sends information/data using TCAP messages.

⑥ ISUP - ISDN user part.

⇒ It is used for basic telephone call set up and call release.

SS7 Networks Architecture

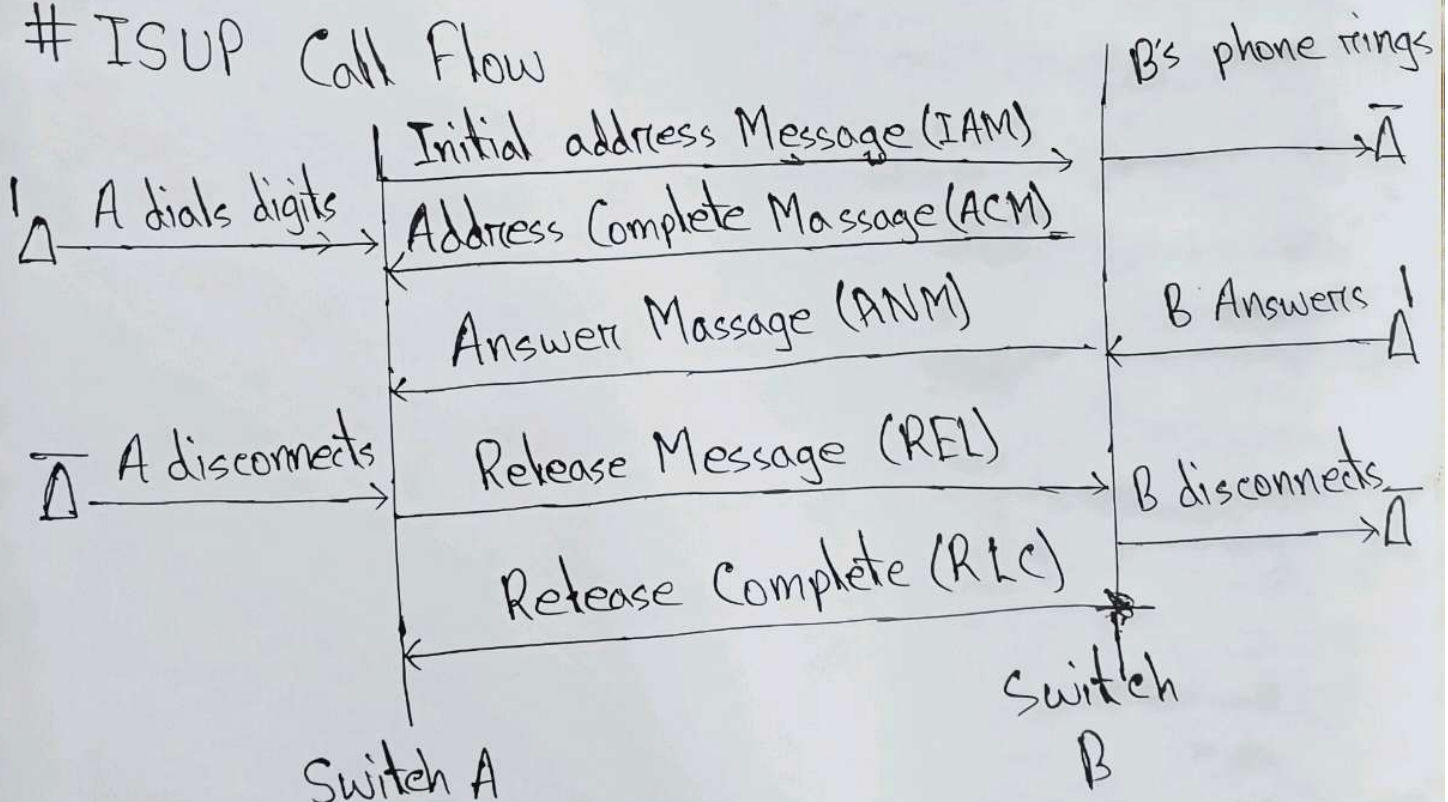
There are 3 main element.

SSP → It is an SS7-capable digital switch.

STP → Routes SS7 messages to the correct destination

SCP → It is a database used mainly for non-call-related signaling.

ISUP Call Flow



Switch A

Switch B

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